CHAPTER SIX

CONCLUSION AND ENHANCEMENT

6.0 Introduction

This chapter concludes the study. It reviews the extent to which the objectives stated in Chapter One were achieved, discusses the weaknesses and problems of the system, and proposes future work.

The study began from the problem described in Chapter One: the property market of Abudwak relies on informal brokers and unstructured listings, prices are set without a reference, and no digital platform exists. Chapter Two reviewed the literature on artificial intelligence, recommender systems and price prediction and showed that no existing platform addresses this market. Chapter Three analysed the requirements and modelled the processes and data of the proposed system, Chapter Four designed its architecture, interface and database, and Chapter Five described its implementation and testing. This final chapter judges the result against the original aims.

6.1 Objective Achievement

The general objective of the project was to design and develop an AI-powered real estate recommendation and price prediction system that gives property seekers, sellers and investors a structured and data-driven alternative to the informal property market. This objective was met in the form of a working web-based system that combines a centralized property database, digital rental and sale workflows and three AI functions. Table 6.1 reviews the four specific objectives.

Table 6.1: Achievement of the Specific Objectives

Specific Objective	What Was Achieved	Status
To develop a hybrid AI recommendation engine using content-based and collaborative filtering	A recommendation engine that combines content-based scoring with preferences learned from each customer's searches, favorites and views. Filtering based on the behaviour of similar users requires more usage data	Partially achieved
To build a price prediction model using Random Forest and Gradient Boosting	A price prediction engine with ridge regression, Random Forest and Gradient Boosting models. It was trained and evaluated on a small sample of listings, not yet on collected Abudwak data	Partially achieved
To integrate a natural language interface for conversational interaction	An AI assistant that answers questions in natural language using current property data from the database	Achieved

Specific Objective	What Was Achieved	Status
To implement a centralized property database and management module	A normalized MySQL database with property management, approval, search, rental, sale, payment and reporting functions	Achieved

In addition to these objectives, the system provides functions that were not stated as objectives in Chapter One but were found necessary during the analysis: digital rental and sale workflows with a defined approving role, the recording of payments with receipts, a central transaction ledger, inquiries and viewing appointments, and an audit log. Table 6.2 relates the research questions of Chapter One to the findings of the study.

Table 6.2: Research Questions and Findings

Research Question	Finding
How can a hybrid recommendation engine match property seekers with suitable listings?	By scoring each available property against stated preferences, with location and budget weighted most, and adjusting the score with the behaviour of the customer; each result is given with reasons
How can a machine learning model generate reliable property valuations in this market?	A regression model trained on structured features performed better than simple baselines on the sample data, but reliable valuation requires a larger set of local transaction data
How can a natural language interface be integrated into the system?	By retrieving the relevant property records from the database and supplying them to a language model, so that answers reflect current listings
To what extent does the centralized database support storage, retrieval and querying?	A normalized relational database supports all the functions of the four actors, and its status rules keep the availability of properties consistent

Contribution of the study. The study contributes a working example of an AI-assisted real estate platform designed for a market in which structured data are scarce. It shows that recommendation, price estimation and conversational assistance can be combined with ordinary property management in one web application built entirely with open-source technologies, and it documents the design in a form that other Somali cities could adapt.

6.2 Weaknesses and Problems of the System

Limited data. The most important weakness is the data available to the AI functions. The price prediction model was trained on a small set of sample listings with asking prices, so its estimates are indicative and cannot yet be regarded as reliable valuations for Abudwak. For the same reason, recommendation based on the behaviour of similar users is limited until the platform has real users.

Payments. The system records and verifies payments by card, bank transfer, cash and mobile money, but it is not connected automatically to the payment providers. A payment made outside the system must be confirmed by the authorized role.

Dependence on external services and connectivity. The AI assistant depends on an external AI service and on an internet connection, and the system as a whole requires connectivity, which is not always stable in the target area.

Language and access. The interface is presented in English, which limits its use by property seekers who are more comfortable in Somali, and the system is available as a web application only.

Testing with users. The system was tested in a local test environment. Its behaviour with a large number of simultaneous users and its acceptance by a wide group of real users remain to be confirmed after deployment.

Table 6.3 summarizes these weaknesses with their effects and the way in which each can be reduced.

Table 6.3: Weaknesses of the System and Their Mitigation

Weakness	Effect	Mitigation
Limited training data	Price estimates are indicative only	Collect local property and transaction data and retrain the models
Few real users	Recommendations rely mainly on stated preferences	Add collaborative filtering as usage data accumulate
Manual payment confirmation	Extra work for managers and administrators	Integrate mobile money and card payment providers
External AI service	The assistant stops when the service or connection fails	Show a clear message and keep search and filters available
English interface	Harder for some property seekers to use	Provide a Somali-language interface and assistant
Local testing only	Performance under load is unknown	Deploy the system and test it with real users

Problems met during development. Several difficulties were met during the project. Reliable property data for Abudwak could not be obtained in a structured form, which is itself evidence of the problem that the study addresses. Keeping the availability of a property consistent across the rental and sale workflows required the status rules to be centralized in one module. Finally, combining the AI functions with the ordinary functions of the system in a single application demanded careful separation of the modules.

6.3 Future Work

The following enhancements are proposed. First, property and transaction data should be collected in Abudwak, in cooperation with local agents, and the price prediction models should be retrained and re-evaluated on these data, including the Random Forest and Gradient Boosting models. Second, collaborative filtering should be added to the recommendation engine when sufficient user activity has accumulated. Third, the system should be integrated with mobile money and card payment providers so that payments are confirmed automatically. Fourth, a Somali-language interface should be provided and the AI assistant should be extended to converse in Somali. Fifth, a mobile application and notifications by SMS would widen access for users with limited connectivity. Finally, the system should be deployed for public use and evaluated with real users over a period of time.

Table 6.4: Proposed Enhancements

Enhancement	Description	Expected Benefit
Local data collection	Gather property and sale records in Abudwak with local agents	Reliable price estimates
Model retraining	Retrain and compare the regression models on the collected data	Higher prediction accuracy
Collaborative filtering	Use the behaviour of similar customers in recommendations	Better personalization
Payment integration	Connect mobile money and card providers	Automatic confirmation of payments
Somali language	Translate the interface and extend the assistant to Somali	Wider access
Mobile application and SMS	Provide a mobile application and SMS notifications	Use with limited connectivity
Public deployment	Host the system and evaluate it with real users	Evidence of real-world usefulness

Recommendations. Real estate agents in Abudwak are encouraged to record their listings and completed transactions in a structured form, since such records are the basis of any reliable price estimation. The university is encouraged to support the collection of a shared local property dataset that future students can use. Future researchers may extend this work to other Somali cities, compare additional prediction models, and study how users in informal markets come to trust the recommendations and estimates of an AI system.

6.4 Chapter Summary

This chapter concluded the study of the AI Agent for Real Estate Recommendation and Price Prediction System. Two of the four specific objectives were achieved in full and two in part,

the difference being due mainly to the limited data available. The weaknesses of the system were identified as limited data, manual confirmation of payments, dependence on external services, language and limited user testing, and future work was proposed to address each of them. The system nevertheless demonstrates that a structured, AI-assisted platform for the property market of Abudwak is feasible.